



# Koneru Lakshmaiah Education Foundation

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

#### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

| TITLE OF THE PROJECT | AI-Powered Dataset Quality Analysis for Machine Learning Pipelines |               |
|----------------------|--------------------------------------------------------------------|---------------|
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| Name of the Guide    |                                                                    |               |

### Abstract

Poor-quality datasets can negatively affect the accuracy, reliability, and performance of Machine Learning models. Detecting missing values, duplicate records, outliers, inconsistent data, incorrect data types, and class imbalance manually can be time-consuming and difficult, especially for large datasets.

The project proposes an AI-powered dataset quality analysis application that automatically analyzes uploaded datasets and identifies common data-quality issues before they are used for Machine Learning. The system provides a centralized dashboard that presents the analysis results, dataset-quality score, detected issues, and recommended preprocessing actions.

Key features of the system include detecting missing values and duplicate records; identifying incorrect or inconsistent data types; detecting potential outliers and anomalous records using Machine Learning techniques; analyzing class distribution to identify class imbalance; performing statistical and correlation analysis; generating an overall dataset-quality score; providing an interactive visual dashboard for dataset analysis; and generating recommendations for improving dataset quality.

The system uses Python, Pandas, NumPy, Scikit-learn, Flask, React, and SQLite for data processing, Machine Learning analysis, backend services, frontend development, and storage. Visualization libraries are used to present dataset statistics, quality metrics, correlations, and detected issues through an interactive dashboard.

The system reduces the time and effort required for manual dataset inspection and helps users identify potential data-quality problems before Machine Learning model development. By providing automated analysis, visual insights, quality scoring, and preprocessing recommendations, the application helps users prepare cleaner and more reliable datasets for Machine Learning tasks.

Signature of the Guide

HOD